FACULTY OF INFORMATION TECHNOLOGY

IT PROJECT 700 | GROUP 55

EMERGEX

Centralized Student Accommodation & Secure Booking Application

PHASE 1 - PROJECT PROPOSAL

Purpose, implementation language, SDLC model and feasibility analysis

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Project lead baseline: Thingo Shezi | *Every member contributes and presents.*

Project Proposal

1. Purpose of the Project

EMERGEX is a centralized South African student accommodation search and secure booking system specifically for Richfield students. It is a response to the Department of Higher Education and Training's constant accommodation shortage report and the practical challenges of comparing trustworthy rooms, timely landlord decisions and transparent booking management. Students can search for verified properties through location, institution, price, whether NSFAS status and Instant Book. You will see rent, deposit, rooms, amenities and visual-evidence categories such as walkthroughs, floor plans and 360-degree tours.

2. Implementation Language

EMERGEX will be implemented using HTML, CSS and JavaScript for the frontend, PHP 8.1 or later for server-side processing, and MySQL or MariaDB through XAMPP for data storage. The booking workflow will clearly display the status of each application in real time. Richfield students can either make an Instant Booking for an eligible property or submit an accommodation request for landlord review. System administrators will verify landlords, property listings and uploaded documentation, while landlords will manage their own properties and booking requests. Students will be able to initiate payment through a recognized South African payment service provider, and EMERGEX will record the payment status returned by that provider. The goal is to provide a centralized and transparent booking process that helps students compare verified accommodation and avoid dealing with unverified listings.

3. SDLC Model

The project will use a Hybrid Model, combining both the Waterfall model and Agile sprints. We will follow the Waterfall model to complete our academic phases step-by-step: Proposal, Planning, Analysis, System Design, and Implementation. Each phase must be reviewed and approved by our supervisor before we move on to the next one. For the actual coding, our six-member team will share roles and work in short, flexible Agile sprints. We will hold weekly check-ins to review our work together and use a simple tracking list to ensure we meet all the project requirements.

4. Feasibility Analysis

A preliminary feasibility analysis indicates that EMERGEX is achievable as a semester academic prototype using the team's available skills, equipment and development time. Feasibility depends on keeping the first release focused on the agreed minimum viable product. Gateway-based payment processing remains in scope.

4.1 Technical Feasibility

The project is technically feasible. The proposed stack uses HTML, CSS and JavaScript for the user interface, PHP 8.1 or later for server-side processing, and MySQL or MariaDB through XAMPP for data storage. These technologies are mature, well documented and able to run on the team's existing Windows laptops. GitHub provides suitable version control and independent branch-based collaboration. The main technical risks are inconsistent local setup, integration defects, payment-gateway integration errors and security weaknesses; these will be reduced through a provider sandbox, a shared setup guide, repeatable tests, code review, backups and controlled merges into the main branch.

4.2 Economic Feasibility

The academic prototype is economically feasible because PHP, XAMPP, MySQL or MariaDB, GitHub and the selected frontend technologies are available without software licence fees. The team will use existing laptops and internet access, so the principal cost is member time and data. Optional production expenses - including hosting, a domain, payment-provider transaction fees, identity verification, security audits, maintenance and customer support - are excluded from the college prototype and would require a separate budget before deployment.

4.3 Operational Feasibility

EMERGEX is operationally feasible for its intended users. Students need a clear search, booking and status-tracking process; landlords need manageable property and booking-request functions; and administrators need verification and audit controls. These responsibilities match the proposed role-based workflow and should require limited training. Key operational risks include incomplete property information, slow landlord responses and a high administrator review workload. The team will mitigate these risks through required fields, verification queues, clear status labels, dummy-data demonstrations and user acceptance testing.

4.4 Schedule Feasibility

The project is feasible within the semester if delivery remains limited to the agreed minimum viable product: verified listings, property search, booking requests, Instant Book, gateway-based payment processing, role-based dashboards and basic reports. The Hybrid SDLC supports this schedule by combining supervisor approval gates for academic phases with short Agile development sprints. Weekly check-ins, task ownership and early testing will be used to control slippage. Escrow, a native mobile application and large-scale national rollout are excluded from Phase 1.

4.5 Legal, Ethical and Security Feasibility

The prototype is conditionally feasible from a legal, ethical and security perspective. POPIA principles require a lawful purpose, minimal data collection, informed handling, access control, secure storage and appropriate retention or deletion. For this reason, demonstrations will use dummy data and must not collect real identity, banking or NSFAS documents. Password hashing, prepared database statements, CSRF protection, input validation, role checks, audit logs and protected uploads form part of the security baseline. Payment will be processed through a regulated South African payment service provider using its hosted or tokenized flow; EMERGEX must not store card or banking credentials. The academic demonstration will use the provider's sandbox or test mode. The application will record provider responses but will not hold, release or manage funds in escrow.

4.6 Overall Feasibility Conclusion

EMERGEX should proceed as an academic prototype. It is technically, economically, operationally and schedule-feasible within the defined scope, using dummy data, provider-based payment processing. Progression to a public production service is feasible only after contracting with a regulated payment provider, confirming legal and POPIA obligations, funding production infrastructure and support, and completing independent security and performance testing.

Table 1. Phase 1 design summary

Design decision	Approved direction
Primary users	Students, Landlords, and Admins.
Core value	Real property pictures, easy booking, clear application status, and secure gateway-based payment processing.
Technology	**Frontend:** HTML, CSS, and JavaScript. **Backend:** PHP. **Database:** MySQL (using MySQL Workbench and XAMPP).
Delivery model	**Hybrid Model** (Using the **Waterfall model** for step-by-step documentation approvals, combined with flexible Agile sprints for coding).
Safety boundary	The project will use dummy data and a payment-provider sandbox or test mode. EMERGEX will not provide escrow or hold and release funds.

Selected sources: DHET student housing review and symposium; NSFAS accommodation guidance; POPIA Act 4 of 2013; PHP and OWASP security guidance. Full URLs appear in the integrated final report.